Environmental Research 286 (2025) 122718



Contents lists available at ScienceDirect

Environmental Research

journal homepage: www.elsevier.com/locate/envres





# Pioneer area construction reduces methylmercury production in mercury-contaminated paddy soils

Piao Ma, Tianrong He *, Deliang Yin **, Jinjuan Li, Xian Zhou

*Key Laboratory of Karst Georesources and Environment (Guizhou University), Ministry of Education, College of Resources and Environmental Engineering, Guizhou University, Guiyang, 550025, China*

ARTICLE INFO

*Keywords:*
Mercury
Hg methylation potential
Rice fields
Planting structure adjustment

ABSTRACT

Paddy fields in mercury (Hg) mining areas are identified as significant sites for Hg methylation, leading to an elevated accumulation of methylmercury (MeHg) in rice. The implementation of pilot zones for soil pollution prevention in China has significantly contributed to pollution control efforts. Its long-term effects on the Hg fate in surface paddy soil remains unclear. This study investigated the impacts of continuous rice cultivation and paddy-dryland conversion on Hg and MeHg in soil in Hg-mining areas over a span of ten years. The results demonstrated a gradual decline in the total Hg (THg) concentrations and a distinct fluctuation in the Hg methylation potential, gauged by the ratio of MeHg to THg, in soil subjected to two different planting patterns over the period from 2012 to 2023. Specifically, Hg methylation potential and MeHg concentrations in soil under continuous rice cultivation witnessed a substantial reduction ($P < 0.05$), falling by 48.6 % and 41.0 %, respectively. In contrast, paddy-to-vegetable conversion exhibited a trend of increase, although there was no statistical significance ($P > 0.05$). The decrease in continuous rice cultivation Hg methylation potential can be attributed not only to the pollution source blocking but also to the rise in molecular inertness of dissolved organic matter and a reduction in soil organic matter. Overall, with the control of pollution, the leaching of irrigation water and the humification of organic matter, Hg pollution in the soil is gradually decreasing, and the Hg methylation potential is also slowly weakening.

## 1. Introduction

Mercury (Hg) is a toxic heavy metal of global concern. In aquatic ecosystems, such as wetlands, inorganic Hg can be methylated into methylmercury (MeHg), a process predominantly mediated by anaerobic Hg-methylating bacteria (Kumar et al., 2024). Given its high neurotoxicity and biomagnification potential within food webs, Hg methylation in soil/sediment is much one of concerns (Aslam et al., 2024; Driscoll et al., 2013).

While the consumption of aquatic products is widely accepted as a primary route for human exposure to MeHg, evidence from the past decade increasingly supports the assertion that rice consumption also serves as a significant pathway (Li et al., 2015). Paddy field constitutes a unique wetland ecosystem, characterized by periodic flooding, which fosters optimal conditions for Hg-methylating bacteria such as sulfate-reducing bacteria. These factors contribute to a substantial increase in MeHg production in soil and its subsequent accumulation in rice (Man et al., 2024; Qiu et al., 2008). Rice primarily absorbs MeHg from soil (Feng et al., 2008). This is assuredly regulated by soil Hg-methylating efficiency, which is further governed by Hg bioavailability, microbial activity, and soil components, including organic matter (OM) and minerals. In paddy soils, physicochemical properties, such as redox potential, powered by flooding-drying cycle, also plays a key role in MeHg production by regulating mineral morphology, OM, and anaerobic Hg-methylating bacteria (Frohne et al., 2012). Specifically, soil OM (SOM) is a more important factor significantly determining MeHg dynamics by regulating Hg bioavailability and microbial activity (Cheng et al., 2023; He et al., 2019). Through its sulfur-containing groups and labile carbon pool, SOM exerts a dual control on MeHg dynamics. Sulfur-containing groups such as thiol can tightly bind Hg and limit its bioavailability, whereas the release of dissolved fraction (DOM) after SOM decomposition would move Hg to elevate its bioavailability and microbial activity (He et al., 2019; Zhang et al., 2024). In recent years, molecular properties of DOM, including

* Corresponding author.
** Corresponding author.
*E-mail addresses:* trhe@gzu.edu.cn (T. He), yindeliang2008@163.com (D. Yin).

https://doi.org/10.1016/j.envres.2025.122718
Received 11 July 2025; Received in revised form 21 August 2025; Accepted 28 August 2025
Available online 17 September 2025
0013-9351/© 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

sources, aromaticity and humic-like components, rather than its concentration, have become the key mechanisms for understanding its interaction with Hg (Zhou et al., 2025). For example, in paddy soils, DOM largely determines Hg-methylating bacteria composition, accounting for 65 %, and its molecular weight acts as a more significant contribution in MeHg production (Pu et al., 2025; Abdelhafiz et al., 2023). Many studies show that paddy soil is facilitate to OM accumulation (Chen et al., 2021; Yang et al., 2024), which may affect MeHg production, while this is less understood.

Historical Hg mining and smelting activities have resulted in the Hg accumulation in agricultural soil. The Wanshan Hg mine in China, recognized as one of the world's largest mines, provides a notable case study for long-term Hg contamination. For example, in surface waters, Hg concentrations ranged from 2.91 to 12,000 ng/L, while in ambient air, the concentrations varied between 17.8 and 1101.8 ng/$m^3$ (Wang et al., 2007). This widespread contamination has led to a substantial accumulation of Hg in paddy soil, 0.5-187.8 mg/kg (Yin et al., 2018), thereby significantly heightening the risk of human exposure to Hg as a result of rice cultivation (Geyman et al., 2025; Feng et al., 2008).

Addressing a paddy system with elevated Hg levels presents substantial challenges for traditional remediation techniques, including chemical stabilization, microbial transformation, and phytoremediation (Boakye et al., 2025; Zhang et al., 2025). Recent efforts have developed a new safe utilization strategy, including cultivation of crops with low Hg accumulation and the transformation of paddy field to other cropping patterns, including peanut, corn, sweet potato, greenhouse, vegetable and landscape forest. Because the ability of Hg bioaccumulation of dryland crops is much lower than that of rice (Meng et al., 2011). This planting transformation is expected to relieve excess Hg exposure risks related to rice consumption. Serving as a demonstration site for China's Soil Pollution Prevention and Control Action Plan, a comprehensive technological framework for the safe utilization of paddy field in the Wanshan Hg mining area has been established. This system integrates source control, pathway interception, and terminal safe utilization to manage Hg-contaminated paddy fields. While the reconstruction of farming patterns presents a viable and safe utilization option, its potential impacts on soil MeHg dynamics remain uncertain.

This study examined the effects of construction of pioneer area on the Hg and MeHg production in surface soil in a Hg mining area. Through a comprehensive analysis of changes in total Hg, MeHg and Hg methylation potential over a span of two decades, we shed light on the importance of source control and SOM evolution.

## 2. Materials and methods

### 2.1. Study area

The Wanshan Hg mine (WHM), located at the eastern margin of the Yunnan-Guizhou Plateau, China. This Hg mine, referred to as China's 'Mercury Capital' due to its vast Hg reserves, is one of the largest Hg mines globally. Long-term Hg mining and smelting activities result in an extremely high Hg accumulation in the local paddy soil. Previous investigations revealed that total mercury (THg) concentrations in paddy soil exhibited a wide range of 0.5 to 187.8 mg/kg, with a mean concentration of 31.0 mg/kg (Yin et al., 2018). This significantly surpasses the risk intervention threshold (4.0 mg/kg), as designated for agricultural soils by the Ministry of Ecology and Environment of China (GB 15618-2018).

### 2.2. Sampling events

During 2012–2023, slag storage, river dredging and planting structure adjustment from paddy fields to dryland types were widespread, as an important strategy of treating Hg-contaminated farmlands. In this study, Wanshan Hg mining area was selected as it is a demonstration site of China's Soil Pollution Prevention and Control Action Plan. Here, '$Soil_R$' represents the paddy soil in 2012. '$Soil_{R\text{-}R}$' represents the sustained rice cultivation. '$Soil_{R\text{-}GAL}$' indicates the transformation from paddy to general tillage dryland, including peanut, sweet potato, and corn. '$Soil_{R\text{-}W}$', '$Soil_{R\text{-}G}$', and '$Soil_{R\text{-}V}$' represents the transformation of paddy to wasteland, greenhouse, and vegetable, respectively.

To ensure comparability, we used GPS data to revisit the paddy sites that were originally investigated in 2012, and collected topsoil (0–5 cm) from 51 sampling sites in August 2023 (similar time in 2012). Among them, there are 19 paddys, 16 general tillage drylands, 8 wastelands, 4 greenhouses lands, and 5 vegetable lands. The spatial distributions of sampling sites are presented in Fig. 1 (Yin et al., 2018). Following the similar protocols in 2012, soil samples were collected from multiple sites within a single paddy field, which were suffering from mixture and homogenization. These samples were transported to the laboratory for subsequent analyses.

### 2.3. Chemical analysis

The soil samples were then scheduled to measure THg and MeHg concentrations, concentration and molecular compositions of dissolved organic matter (DOM), and Soil Organic Matter (SOM content.

We have employed a widely accepted quantitative methodology for the determination of THg and MeHg (Liang et al., 1994, 1996). For THg, soil samples underwent a standardized treatment process involving acid digestion, followed by BrCl oxidation, $SnCl_2$ reduction, and gold trapping, and subsequent detection using cold vapor atomic fluorescence spectrometry (CVAFS; Brooks Rand Labs, USA). For MeHg, the soil samples were extracted by $HNO_3/CuSO_4$ solution and $CH_2Cl_2$, followed by ethylation, Tenax trapping, gas chromatography separation, high-temperature pyrolysis, and subsequent detection using CVAFS.

The SOM content in the soil samples was quantified utilizing the hydration calorimetry method, employing a precision-controlled water bath (Huang, 1994).

The soil DOM was extracted with a soil and water ratio of 1:10 (w/v). The mixed liquid was subjected to continuous shaking at 220 rpm for 24 h, followed by high-speed centrifugation at 4000 rpm for 20 min. The supernatant was then filtered through 0.45 μm filters to determine DOC concentration using a TOC analyzer (Vario TOC, Elementar, Hessen, Germany) and fluorescence characteristics using a spectrofluorometer (AquaLog, Horiba, Kyoto, Japan), and the index calculation is shown in Table S1 in the SI.

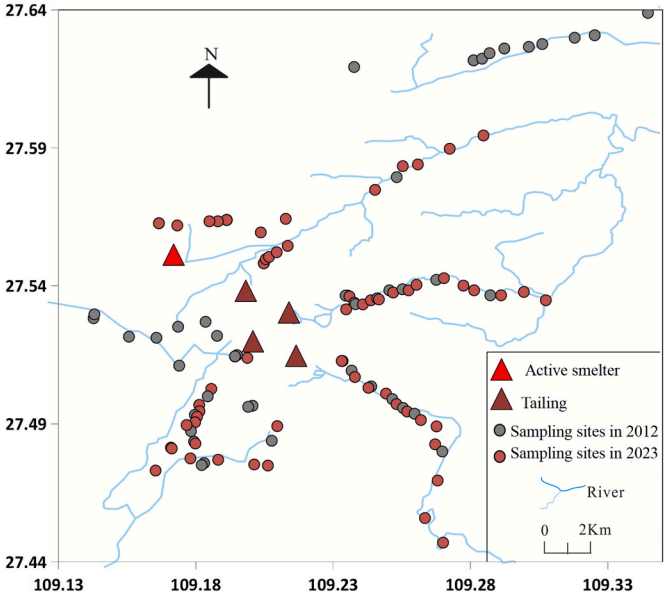



**Fig. 1.** Distribution of the sampling sites in 2012 and 2023.

### 2.4. Quality assurance and quality control

Quality assurance and quality control (QA/QC) were performed by duplicate sample analysis and certified reference material (CRM). The QA/QCs of THg and MeHg in soil were conducted using GSS-5 (GBW07405, Institute of Geophysical and Geochemical Exploration, Chinese Academy of Geological Sciences) and ERM-CC580 (Estuarine sediment, Joint Research Centre, European Commission), respectively. Recovery rates of CRMs in this study ranged from 80 % to 120 %, and the relative standard deviations of replicate analyses were below 12 %.

### 2.5. Data analysis

Statistical analysis and data processing were conducted using SPSS 27.0, GetData 2.20, Microsoft Excel 2019, and Origin 2021. Non-parametric statistical comparisons were performed using the Wilcoxon signed-rank test, following standardized statistical protocols.

The MATLAB 2016b combined with the EfC platform (version v1.0; https://www.nomresearch.cn/) was used to perform parallel factor analysis (PARAFAC). This approach successfully identified three predominant fluorescent components, including two humic-like components (C1 and C2) and one protein-like component (C3). After fingerprint database comparison, C1 is classified as a mixture of traditional terrestrial humic-like peaks A and C (Coble, 1996); C2 is similar to terrestrial quinone-like components; C3 represents freshly produced tryptophan-like components (Yamashita et al., 2010, 2011). Quantitative assessment of these components was achieved through maximum fluorescence intensity (Fmax).

## 3. Results and discussion

### 3.1. Inter-annual variation of THg

The currently accessible data, combined with our field investigation in 2023, revealed a clear temporal pattern of THg in paddy soil (Zhang et al., 2010; Yin et al., 2014; Zhu, 2018; Gao et al., 2021; Horvat et al., 2003). During 2003–2023, THg concentration in soil over recent years showed a slow decreasing trend, although this decrease was only statistically significant between 2003 and 2023 ($P < 0.05$, Fig. 2a). A comparison of THg concentration in soil at the same sampling site between 2023 and 2012 revealed that THg concentration in dryland soil also showed a general decreasing trend (Fig. 2b). This decline may be attributed to the diminishment of pollution sources, coupled with the water erosion and bioaccumulation of Hg in paddy soil. The cessation of mining and smelting activities, the Hg concentration in the atmosphere has decreased from 2006 to 2024 by 48 % (Wang et al., 2006; Cao et al., 2024) (Fig. 3a). Moreover, the residual tailings from Hg mining operations constitute a substantial source of Hg in paddy fields (Dai et al., 2013; Yan et al., 2025). Recent efforts have been undertaken to seal these tailings, in an attempt to inhibit the discharge of Hg into nearby rivers (Yan et al., 2025). As a result of these measures, Hg concentration in river water in the study area decreased from 2016 to 2021 by 62.3 % (Wu, 2021; Xia et al., 2016) (Fig. 3b). On the other hand, water erosion and leaching may have reduced the Hg pool in surface soil. The Wanshan Hg mining area has a relatively high slope (45°), facilitating the Hg erosion (Dai et al., 2013). Previous study showed that paddy system was a significant source of Hg for downstream lakes and rivers via runoff (Zhong et al., 2024). In addition, the uptake of Hg by rice, a highly bioaccumulation crop, can also remove a certain amount of Hg from the soil (Chen et al., 2025).

### 3.2. Inter-annual variability of Hg methylation potential

We further compared the soil MeHg from the period spanning 2003 to 2023 (Gao et al., 2021; Yin et al., 2014; Horvat et al., 2003; Li et al., 2016). As depicted in Fig. 4a, an obvious decrease in MeHg concentration in soil was found over time. Specifically, we noted a statistically significant decline ($P < 0.01$), characterized by a marked reduction of 41 % between 2012 and 2023. The correlation between THg and MeHg indicates a significant relationship in paddy soil from 2012. However, this relationship diminishes in paddy soil from 2023 and after conversion to dryland soil (Fig. 5). This implies that there has been a decoupling between Hg and MeHg in surface soil after the construction of pioneer area. The capability of total Hg to forecast MeHg variation continued to decrease.

In this study, the ratio of MeHg to THg (%MeHg) was employed to represent the methylation potential. Fig. 4b depicts the fluctuation in % MeHg across a span of 20 years. The %MeHg value, which is less than 0.1 %, indicates a relatively low Hg methylation level. This finding aligns with the previous report (Yin et al., 2018). Time-series analysis

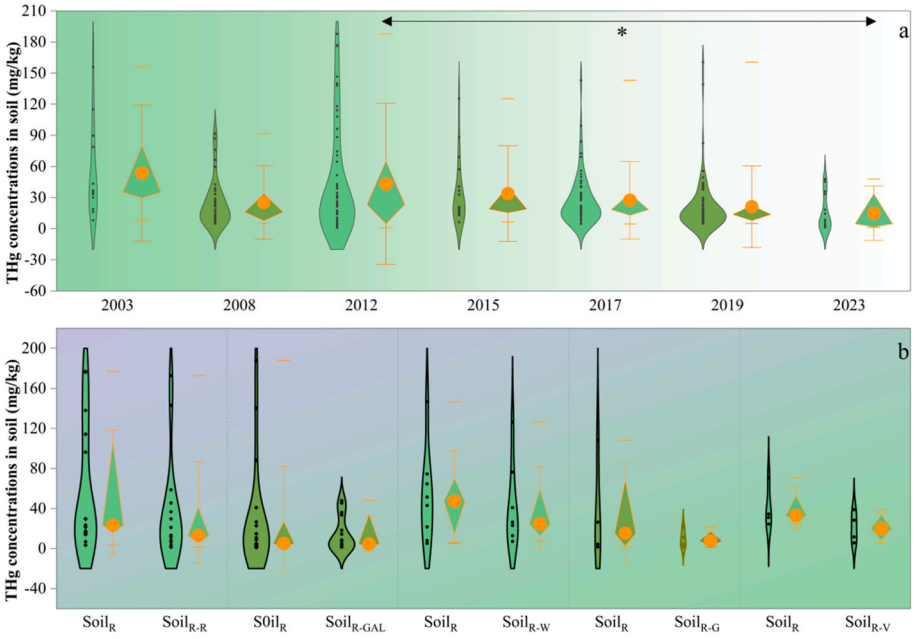



**Fig. 2.** Dynamics of THg concentrations in surface soils (a) and the effects of planting structure adjustments on THg concentrations (b).

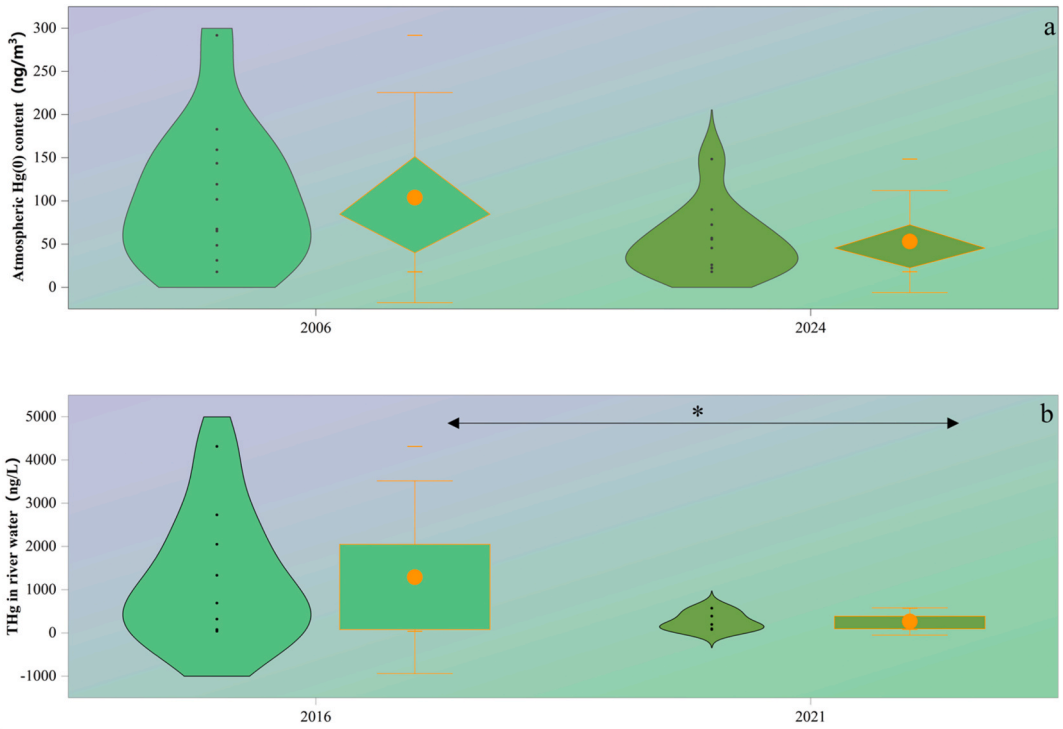



**Fig. 3.** Variation of $Hg^0$ in atmosphere(a) and THg in surface water (b) n Wanshan Hg mining area.

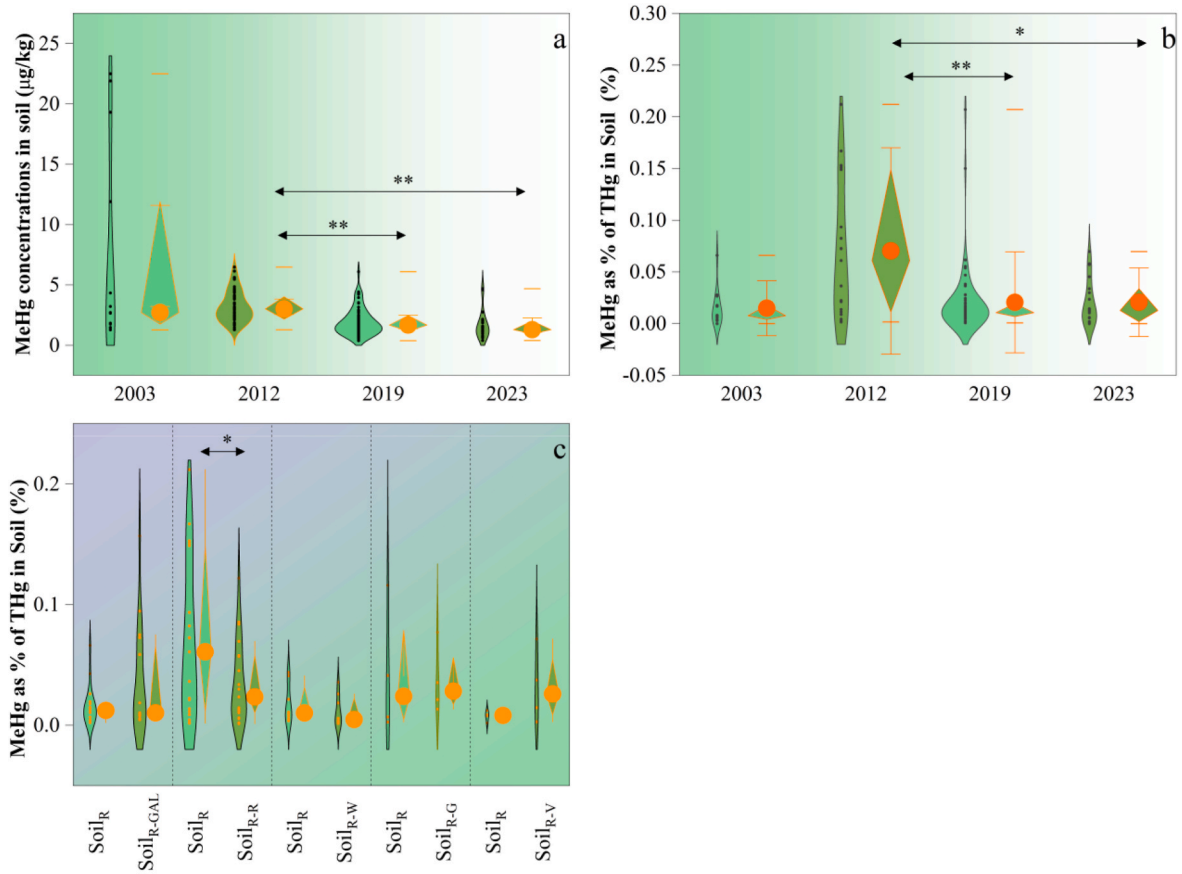



**Fig. 4.** Dynamic of MeHg concentrations and Hg methylation potential (%MeHg) in surface soils over time (a,b) and Hg methylation potential (%MeHg) of the impacts of planting structure adjustments (c).

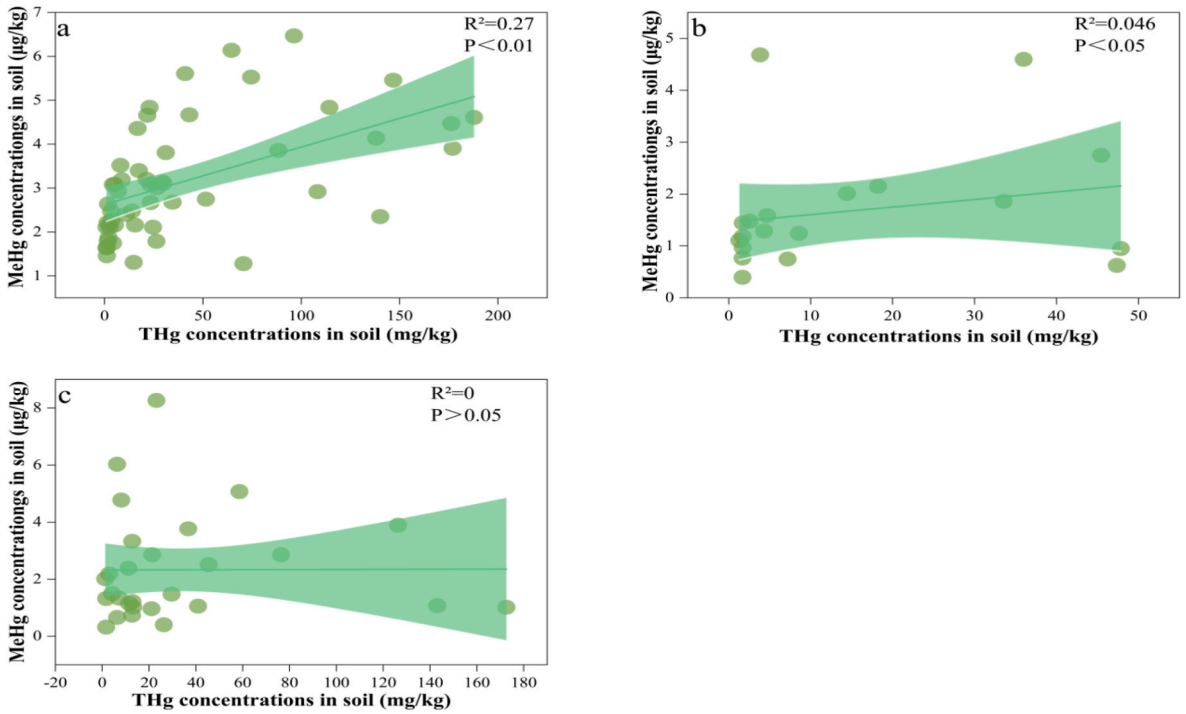



**Fig. 5.** Correlation between THg and MeHg in paddy soil in 2012 (a), dry land in 2023 (b), and paddy soil in 2023 (c).

revealed a gradual decline in %MeHg over time, suggesting a diminished Hg methylation potential in surface soil. Fig. 4c is dedicated to reveal the %MeHg variation at the same observation site suffering from planting structure adjustment campaign. Compared with %MeHg of the paddy field in 2012, continuous rice cultivation ($Soil_{R\text{-}R}$) significantly reduced soil %MeHg level ($P < 0.05$). In contrast, we found that $Soil_{R\text{-}V}$ exhibited a non-significant increase ($P > 0.05$). The transitioning to other planting patterns ($Soil_{R\text{-}G}$, $Soil_{R\text{-}W}$, and $Soil_{R\text{-}GAL}$) resulted in no significant impacts on the %MeHg, indicating that other planting structure adjustments did not markedly modify Hg methylation potential ($P > 0.05$).

The Hg methylation potential in soil is related to factors such as available Hg content, redox potential, organic matter, and pH in soil (Wang et al., 2021; Huang et al., 2025). Among all the adjustments in planting structure, we can observe a slight increase in Hg methylation potential in soil that has been converted into vegetable fields. This phenomenon has been reported in previous research (Qiu, 2005). Although we still cannot provide a clear explanation in terms of mechanism, according to the vegetable planting habits, frequent irrigation with liquid farmyard manure may result in a prolonged state of semi-wet conditions and elevated DOM input. As a result, this soil exhibits a higher potential for Hg methylation. A further study in the future work is needed.

Compared with the soil that has been adjusted to other planting patterns, it is observed that the soil which has been continuously used for rice cultivation exhibits a reduced methylation potential. It may be related to the reduction of available Hg in soil and the temporal changes of SOM. As mentioned in 3.1, the THg concentration in soil is gradually decreased due to factors such as the diminishment of pollution sources, water flushing, and plant absorption. The diminishment of pollution sources has consequently led to a reduction of newly deposited Hg. The freshly deposited atmospheric Hg exhibits substantially higher activity compared to aged Hg in soil (Luo et al., 2024; Chen et al., 2024). These alterations reduce the methylated Hg species in surface soil. Compared with other planting patterns, the frequent wet-dry alternation and ploughing in paddy fields makes soil Hg more prone to soaking and erosion, especially the dissolved and available Hg. Previous study has found that paddy field is one of important sources of total dissolved solids, indicating the non-negligible contribution of irrigation (Ruark et al., 2010). This results in a lower Hg methylation potential in the soil that has been continuously planted with rice. In addition, the interannual variation in the Hg methylation potential may also be related to the continuous input, mineralization and humification of SOM, which will be discussed in 3.3.

### 3.3. *Influence of temporal changes of SOM on the Hg methylation potential*

Content and properties of SOM are key regulators of MeHg production by fueling Hg methylating bacteria and regulating Hg bioavailability (Pu et al., 2025; Abdelhafiz et al., 2023). Therefore, the changes in SOM over time may also contribute to the variation of Hg methylation potential in soil that have been continuously cultivated with rice, or transformed to other planting patterns.

Fig. 6 shows the differences in OM content, DOC content and their molecular properties in soil for continuous rice cultivation. Compared with 2012, the DOC in the soil increased (Fig. 6a) while the SOM decreased in 2023 (Fig. 6c). Meanwhile, the humification level of DOM was enhanced (Fig. 6g). Wetting-drying cycle can stimulate aggregate breakdown and irrigation erosion, resulting in a loss of SOM (Liu et al., 2024). This may further elevate the DOM release in soil, while promote its humification and aromaticity by enhancing microbial activity (Eckley et al., 2017; Liu et al., 2025). This suggests that the SOM in paddy field is undergoing a gradual mineralization and humification. The increase in DOM and decline in SOM indicate that the retention of Hg and MeHg diminishes, whereas irrigation erosion and mobility are likely enhanced. Although DOM (Fig. 6a) and its colored DOM fraction (Fig. 6b) are elevated, its labile protein-like fractions showed a decrease from 2012 to 2023, while its humus-like fractions showed a contrast pattern (Fig. 6c). In addition, molecular weight (Fig. 6f) and humification (Fig. 6g) of DOM were increased ($P < 0.05$). This implies that, in recent years, soil DOM has demonstrated the elevated complexity and stability. Previous study finds that high concentrations of highly humified DOM hamper Hg methylation by decreasing Hg bioavailability and bacterial activity (Bravo et al., 2018). We also establish a highly significant negative

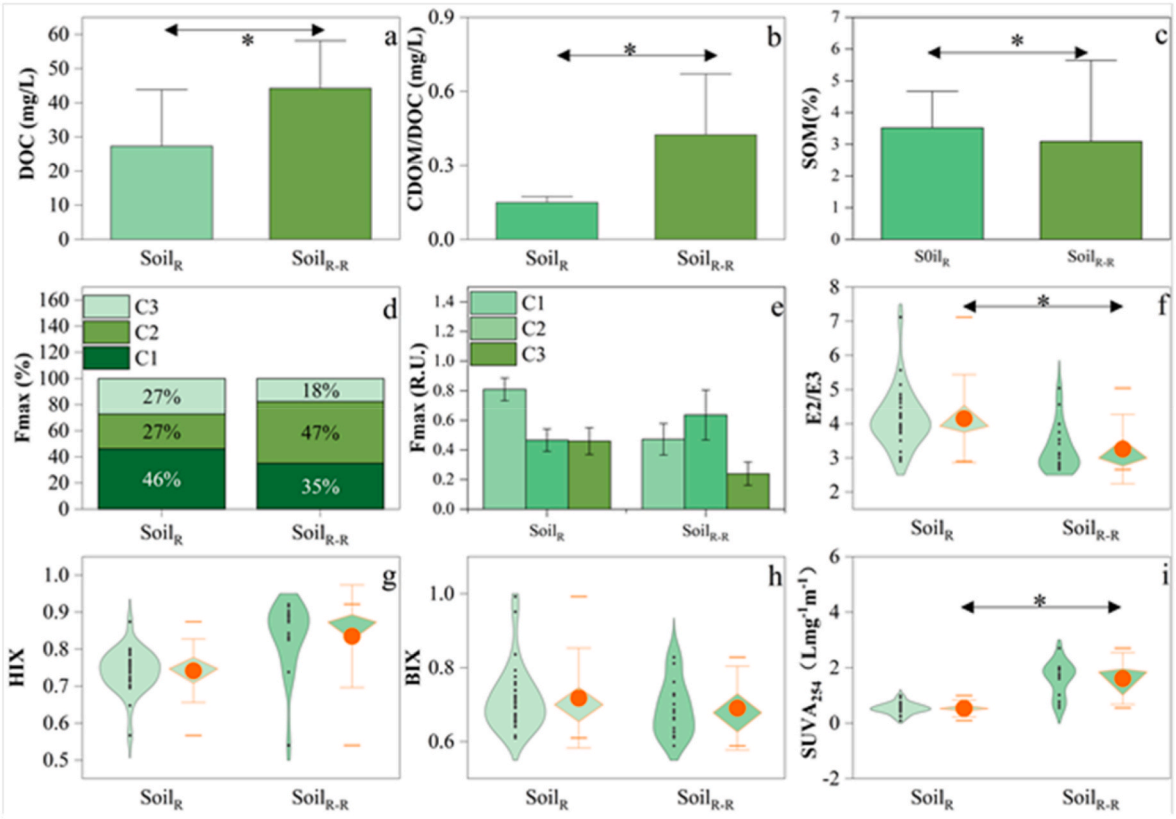



**Fig. 6.** Spectral characteristics of soil DOM, including the ratio of DOC concentrations (a), CDOM to DOC(b), SOM concentrations (c), maximum fluorescence intensity (Fmax) of components (d,e), E2/E3(f), HIX(g), BIX(h) and $SUVA_{254}$(i), under different planting patterns. (* represents significant differences at 0.05 level).

correlation ($P < 0.01$) between the humic-like components (C2) and the %MeHg. This further indicates that as SOM continuously humifies, the Hg methylation potential gradually decreases. Inert DOM components such as humic moieties with highly reduced sulfur (i.e., sulfides or thiols) result in a preferred Hg affinity and reduce the microbial Hg bioavailability by sequestering it in stable organic–Hg complexes (He et al., 2019; Wang et al., 2022). Consequently, this will partially result in the decrease of Hg methylation potential in soil, further indicating that as SOM continuously humifies, the Hg methylation potential gradually decreases.

In contrast, after being transformed into dryland, the SOM also decreases, while DOM properties such as humification, aromaticity, DOC and molecular weight increase from 2012 to 2023 (Fig. S1 and SI). However, compared to $Soil_{R\text{-}R}$, $Soil_{R\text{-}D}$ has larger deceases of SOM (21.5 % vs 5.0 %), larger increases of aromaticity (364.0 % vs 201.0 %), while lower increases of DOC (53.9 % vs 89 %) and molecular weight (13.8 % vs 21.5 %). Specially, they have no-significant difference in humification (17.0 % vs 13.0 %, $P > 0.05$). These DOM properties indicate that $Soil_{R\text{-}D}$ has relatively lower molecular weight but higher aromaticity than $Soil_{R\text{-}R}$. Although the non-flooding condition of dryland is conductive for SOM oxidation and not for anaerobic microbial Hg methylation, elevated DOM aromaticity which has been proved to facilitate desorption of particle-bound Hg may increase Hg bioavailability and enhance Hg methylation (Krause et al., 2024; Jiang et al., 2018). This may balance the potential inhibitory effect of enhanced DOM humification and aerobic condition on Hg methylation, ultimately resulting in no obvious change of MeHg in $Soil_{R\text{-}D}$.

## 4. Conclusion and limitations

This study investigates the effects of establishing soil pollution prevention pilot zones in a Hg mining area on the variations of Hg species in paddy soil. The result has shown that, although paddy soil is a suitable site for Hg methylation, continuous rice cultivation over the past two decades under pioneer-area interventions has led to a decline in Hg methylation potential in surface soil. This could be attributed to the fact that water erosion may have diminished the Hg pool in the surface soil. Concurrently, the decline in exogenous Hg inputs, coupled with the stabilization of SOM in paddy soil, may also contribute to this reduction. Nevertheless, converting paddy fields into vegetables exhibites a non-significant increase, needing a further attention. The transitioning to other planting patterns ($Soil_{R\text{-}G}$, $Soil_{R\text{-}W}$, and $Soil_{R\text{-}GAL}$) results in no significant impacts on the Hg methylating capacity. It needs to be emphasized that this study has several limitations. Confined to the difficulty of obtaining historical data, we cannot capture the evolutionary signature of Hg methylating bacteria. In addition, soil properties such as sulfate positively regulating Hg bioavailability and Hg methylating bacteria were not collected. These informations may provide a more comprehensive evaluation considering the interannual variations of Hg methylation potential, an area that warrants further exploration in future studies.

## CRediT authorship contribution statement

**Piao Ma:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Tianrong He:** Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Deliang Yin:** Resources, Methodology, Data curation, Conceptualization. **Jinjuan Li:** Writing – review & editing, Conceptualization. **Xian Zhou:** Visualization, Data curation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgment

This work was supported by the Natural Science Foundation of China (Nos.42477268, 22166009, and 42267032), Natural Science Research Project of Guizhou Provincial Department of Education (QianjiaoJi [2024] 319), Guizhou Provincial Basic Research Program (Natural Science) (No. Qiankehejichu-ZD [2025]080) and High-Level Talent Training Program in Guizhou Province (GCC [2023]045).

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envres.2025.122718.

## Data availability

Data will be made available on request.

## References

Abdelhafiz, M.A., Liu, J., Jiang, T., Pu, Q., Aslam, M.W., Zhang, K., Meng, B., Feng, X., 2023. DOM influences Hg methylation in paddy soils across a Hg contamination gradient. Environ. Pollut. 322.

Aslam, M.W., Meng, B., Ali, W., Abrar, M.M., Abdelhafiz, M.A., Feng, X., 2024. Low mercury risks in paddy soils across the Pakistan. Sci. Total Environ. 943.

Boakye, K.O., Dodd, M., Asante, M.D., Logah, V., Darko, G., 2025. Biochar amendment in remediation of heavy metals in paddy soil: a Case Study in Nobewam, Ghana, 9 (2), 38.

Bravo, A.G., Kothawala, D.N., Attermeyer, K., Tessier, E., Bodmer, P., Ledesma, J.U., Audet, J., Pere Casas Ruiz, J., Catalan, N., Cauvy Fraunie, S., Colls, M., Deininger, A., Evtimova, V.V., Fonvielle, J.A., Fuss, T., Gilbert, P., Ortega, S.H., Liu, L., Mendoza Lera, C., Monteiro, J., Mor, J.R., Nagler, M., Niedrist, G.H., Nydahl, A.C., Pastor, A., Pegg, J., Roberts, C.G., Pilotto, F., Portela, A.P., Gonzalez Quijano, C.R., Romero, F., Rulik, M., Amouroux, D., 2018. The interplay between total mercury, methylmercury and dissolved organic matter in fluvial systems: a latitudinal study across Europe. Water Res. 144, 172–182.

Cao, Q., Hu, H., Yuan, W., Huang, J., Fu, X., Feng, X., 2024. Isotope-Based characterization of soil elemental mercury emissions from historical mercury mining areas: driving pathways and relative contributions. Environ. Sci. Technol. 58 (38), 16824–16832.

Chen, Q., Wu, Q., Cui, Y., Wang, S., 2024. Mercury records from natural archives reveal ecosystem responses to changing atmospheric deposition. Natl. Sci. Rev. 11 (12).

Chen, X., Hu, Y., Xia, Y., Zheng, S., Ma, C., Rui, Y., He, H., Huang, D., Zhang, Z., Ge, T., Wu, J., Guggenberger, G., Kuzyakov, Y., Su, Y., 2021. Contrasting pathways of carbon sequestration in paddy and upland soils. Glob. Change Biol. 27 (11), 2478–2490.

Chen, X., Yu, B., Wang, X., Zhu, R., Zhang, L., 2025. The uptake and in-vivo migration of Hg by plants: a critical review. Rev. Environ. Sci. Biotechnol. 24 (1), 145–165.

Cheng, Z., He, T., Yin, D., Tian, X., Ran, S., Zhou, X., 2023. Effects of composted agricultural organic materials on mercury methylation in paddy soil and mercury enrichment in rice. Bull. Environ. Contam. Toxicol. 110 (1).

Coble, P.G.J.M.C., 1996. Characterization of marine and terrestrial DOM in seawater using excitation-emission matrix spectroscopy. Mar. Chem. 51 (4), 325–346.

Dai, Z., Feng, X., Zhang, C., Shang, L., Qiu, G., 2013. Assessment of mercury erosion by surface water in Wanshan mercury mining area. Environ. Res. 125, 2–11.

Driscoll, C.T., Mason, R.P., Chan, H.M., Jacob, D.J., Pirrone, N., 2013. Mercury as a global pollutant: sources, pathways, and effects. Environ. Sci. Technol. 47 (10), 4967–4983.

Eckley, C.S., Luxton, T.P., Goetz, J., McKernan, J., 2017. Water-level fluctuations influence sediment porewater chemistry and methylmercury production in a flood-control reservoir. Environ. Pollut. 222, 32–41.

Feng, X., Li, P., Qiu, G., Wang, S., Li, G., Shang, L., Meng, B., Jiang, H., Bai, W., Li, Z., Fu, X., 2008. Human exposure to methylmercury through rice intake in mercury mining areas, guizhou province, China. Environ. Sci. Technol. 42 (1), 326–332.

Frohne, T., Rinklebe, J., Langer, U., Du Laing, G., Mothes, S., Wennrich, R., 2012. Biogeochemical factors affecting mercury methylation rate in two contaminated floodplain soils. Biogeosciences 9 (1), 493–507.

Gao, L., Mao, K., Zhang, W., Cui, Z., Lu, B., Huang, G., Zhang, J., Feng, X., Zhang, H., Shang, L., 2021. Distribution and Pollution characteristics of soil mercury in rice paddies in wanshan mercury mining area, Guizhou, China. Bull. China Soc. Mineral Petrol. Geochem. 40 (1), 148–154.

Geyman, B.M., Streets, D.G., Olson, C.I., Thackray, C.P., Olson, C.L., Schaefer, K., Krabbenhoft, D.P., Sunderland, E.M., 2025. Cumulative anthropogenic impacts of past and future emissions and releases on the global mercury cycle. Environ. Sci. Technol.

He, M., Tian, L., Braaten, H.F.V., Wu, Q., Luo, J., Cai, L., Meng, J., Lin, Y., 2019. Mercury-Organic matter interactions in soils and sediments: angel or Devil? Bull. Environ. Contam. Toxicol. 102 (5), 621–627.

Horvat, M., Nolde, N., Fajon, V., Jereb, V., Logar, M., Lojen, S., Jacimovic, R., Falnoga, I., Liya, Q., Faganeli, J., Drobne, D., 2003. Total mercury, methylmercury and selenium in mercury polluted areas in the province Guizhou, China. Sci. Total Environ. 304 (1–3), 231–256.

Huang, G.C.M.L.D., 1994. Constant-boiling water bath hydration calorimetry for soil organic matter determination. Guangxi Agricultural Science (3), 125–129.

Huang, Z., Zheng, D., Gao, Q., Ma, Z., Sun, Y., 2025. Study on factors influencing mercury methylation in Shenyang Qixing wetland under different seasonal conditions. Water, Air, Soil Pollut. 236 (7).

Jiang, T., Bravo, A.G., Skyllberg, U., Bjorn, E., Wang, D., Yan, H., Green, N.W., 2018. Influence of dissolved organic matter (DOM) characteristics on dissolved mercury (Hg) species composition in sediment porewater of lakes from southwest China. Water Res. 146, 146–158.

Krause, V.M., Baldwin, A.K., Peterson, B.D., Krabbenhoft, D.P., Janssen, S.E., Willacker, J.J., Eagles Smith, C.A., Poulin, B.A., 2024. Riparian methylmercury production increases riverine mercury flux and food web concentrations. Environ. Sci. Technol.

Kumar, A., Kumar, V., Bakshi, P., Parihar, R.D., Radziemska, M., Kumar, R., 2024. Mercury in the natural environment: biogeochemical cycles and associated health risks. J. Geochem. Explor. 267.

Li, P., Feng, X., Chan, H., Zhang, X., Du, B., 2015. Human body burden and dietary methylmercury intake: the relationship in a Rice-Consuming population. Environ. Sci. Technol. 49 (16), 9682–9689.

Li, R., Xu, X., Xu, Z., Ao, M., Shang, L., Qiu, G., Tang, S., 2016. Total mercury and methylmercury in rice and exposure risks in a typical mercury mining area watershed in Guizhou Province. Research of Environmental Sciences 29 (12), 1829–1839.

Liang, L., Horvat, M., Bloom, N.S., 1994. An improved speciation method for mercury by GC/CVAFS after aqueous phase ethylation and room temperature precollection. Talanta 41 (3), 371–379.

Liang, L., Horvat, M., Cernichiari, E., Gelein, B., Balogh, S., 1996. Simple solvent extraction technique for elimination of matrix interferences in the determination of methylmercury in environmental and biological samples by ethylation-gas chromatography-cold vapor atomic fluorescence spectrometry. Talanta 43 (11), 1883–1888.

Liu, S., Huang, J., He, W., Shi, L., Zhang, W., Li, E., Hu, J., Zhang, C., Pang, H., 2025. Insights into effects of drying-wetting cycles on dissolved organic matter and Cd bioavailability in riparian sediments amended with microplastics. Environ. Res. 271.

Liu, S., Huang, X.J., Gan, L., Zhang, Z.B., Dong, Y., Peng, X.H., 2024. Drying-wetting cycles affect soil structure by impacting soil aggregate transformations and soil organic carbon fractions. Catena 243.

Luo, K., Yuan, W., Lu, Z., Xiong, Z., Lin, C., Wang, X., Feng, X., 2024. Unveiling the sources and transfer of mercury in forest bird food chains using techniques of vivo-nest video recording and stable isotopes. Environ. Sci. Technol. 58 (13), 6007–6018.

Man, Y., Wang, B., Wang, J., Cai, K., Rinklebe, J., Zhang, L., Feng, X., 2024. New insights into MeHg accumulation in rice (Oryza sativa L.): evidence from cysteine. Environ. Sci. Technol. 58 (13), 5942–5951.

Meng, B., Feng, X., Qiu, G., Liang, P., Li, P., Chen, C., Shang, L., 2011. The Process of methylmercury accumulation in rice (Oryza sativa). Environ. Sci. Technol. 45 (7), 2711–2717.

Pu, Q., Meng, B., Huang, J., Zhang, K., Liu, J., Liu, Y., Abdelhafiz, M.A., Feng, X., 2025. Dissolved organic matter fosters core mercury-methylating microbiomes for methylmercury production in paddy soils. Biogeosciences 22 (6), 1543–1556.

Qiu, G., 2005. Environmental Geochemistry of Mercury in Typical Mercury Mining Areas of Guizhou Province. Institute of Geochemistry, Chinese Academy of Sciences.

Qiu, G., Feng, X., Li, P., Wang, S., Li, G., Shang, L., Fu, X., 2008. Methylmercury accumulation in rice (Oryza sativa) grown at abandoned mercury mines in Guizhou, China. J. Agric. Food Chem. 56 (7), 2465–2468.

Ruark, M.D., Linquist, B.A., Six, J., van Kessel, C., Greer, C.A., Mutters, R.G., Hill, J.E., 2010. Seasonal losses of dissolved organic carbon and total dissolved solids from rice production systems in Northern California. J. Environ. Qual. 39 (1), 304–313.

Wang, J., Shaheen, S.M., Jing, M., Anderson, C.W.N., Swertz, A.C., Wang, S., Feng, X., Rinklebe, J., 2021. Mobilization, Methylation, and Demethylation of mercury in a paddy soil under systematic redox changes. Environ. Sci. Technol. 55 (14), 10133–10141.

Wang, S., Feng, X., Qiu, G., fu, X., 2006. Study of mercury exchange fluxes at the surface-atmosphere interface in the Wanshan mercury mine area. Environ. Sci. J. Integr. Environ. Res. 27 (8), 1487–1494.

Wang, S., Feng, X., Qiu, G., Fu, X., Wei, Z., 2007. Characteristics of mercury exchange flux between soil and air in the heavily air-polluted area, eastern Guizhou, China. Atmos. Environ. 41 (27), 5584–5594.

Wang, Y., Liu, J., Van, L.N., Tian, S., Zhang, S., Wang, D., Jiang, T., 2022. Binding strength of mercury (II) to different dissolved organic matter: the roles of DOM properties and sources. Sci. Total Environ. 807.

Wu, G., 2021. Characterization and Mechanism of Mercury Transfer in Aquatic Food Chain in Wanshan Mercury Mining Area. Bachelor's Degree. Henan University of Science and Technology, Guizhou Province, China.

Xia, J., Hu, P., Wang, J., Zhang, H., Xinbin, F., 2016. Characterization of mercury pollution in Tongren mercury mining area in Guizhou Province. Asian J. Ecotoxicol. 11 (1), 231–238.
Yamashita, Y., Kloeppel, B.D., Knoepp, J., Zausen, G.L., Jaffe, R., 2011. Effects of watershed history on dissolved organic matter characteristics in headwater streams. Ecosystems 14 (7), 1110–1122.
Yamashita, Y., Maie, N., Briceno, H., Jaffe, R., 2010. Optical characterization of dissolved organic matter in tropical rivers of the Guayana Shield, Venezuela. Journal of Geophysical Research-Biogeosciences 115.
Yan, J., Li, R., Wang, C., Yang, S., Shao, M., Zhang, L., Li, P., Feng, X., 2025. Transport and transformation of colloidal and particulate mercury in contaminated watershed. Water Res. 278.
Yang, X., Song, Z., Van Zwieten, L., Guo, L., Chen, J., Luo, Z., Wang, Y., Luo, Y., Wang, Z., Wang, W., Wang, J., Wang, Y., Liu, C.Q., Wang, H., 2024. Significant accrual of soil organic carbon through long-term rice cultivation in paddy fields in China. Glob. Change Biol. 30 (3).
Yin, D., He, T., An, Y., Luo, G., Xu, Y., Chen, J., 2014. Study on the distribution characteristics and influencing factors of mercury content in Wanshan rice. Journal of Agro-Environment Science (6), 1082–1088.
Yin, D., He, T., Yin, R., Zeng, L., 2018. Effects of soil properties on production and bioaccumulation of methylmercury in rice paddies at a mercury mining area, China. J. Environ. Sci. 68, 194–205.
Zhang, H., Feng, X., Larssen, T., Shang, L., Vogt, R.D., Rothenberg, S.E., Li, P., Zhang, H., Lin, Y., 2010. Fractionation, distribution and transport of mercury in rivers and tributaries around Wanshan Hg mining district, Guizhou province, southwestern China: Part 1-Total mercury. Appl. Geochem. 25 (5), 633–641.
Zhang, S., Xia, M., Pan, Z., Wang, J., Yin, Y., Lv, J., Hu, L., Shi, J., Jiang, T., Wang, D., 2024. Soil organic matter degradation and methylmercury dynamics in Hg-contaminated soils: relationships and driving factors. J. Environ. Manag. 356, 120432.
Zhang, Y., Yang, Y., Yu, X., Yang, S., Sun, J., Sun, T., Zhang, X., Ye, H., Sun, Y., Li, X., 2025. Biocurrent regulates the mercury transformation in paddy soil. J. Clean. Prod. 489.
Zhong, H., Zhou, H., Li, Y., Li, C., Tsui, M.T.-K., Mitchell, C.P.J., Zhou, Y., Yang, Y., Chen, L., Ren, H., Tang, W., 2024. Methylmercury photodegradation in paddy water: an overlooked process mitigating methylmercury risks. Water Res. 253.
Zhou, X., He, T., Yin, Y., Jiang, T., Wu, P., Liu, J., Wang, Y., Yin, D., Liu, E., Ma, S., Xie, Q., 2025. Elevated methylmercury production in seasonally inundated sediments: insights from DOM molecular composition. J. Hazard Mater. 487.
Zhu, Z., 2018. Research on the Method of Analyzing the Biologically Effective State of Mercury-Contaminated Soil in Karst Area and Remediation Technology. Bachelor's Degree. University of Chinese Academy of Sciences.